

Deliverable 2: High Level Design and Trade-off Analysis

Deliverable 2 demonstrates that the student can systematically explore engineering solutions, evaluate alternatives using trade-off analysis, and define a defensible high-level system architecture composed of interacting subsystems.

Purpose of this deliverable:

Deliverable 2 marks the transition from problem understanding (D1) to solution exploration. Where Deliverable 1 proved that the problem is a complex engineering problem, Deliverable 2 demonstrates that the student can think like an engineer and explore defensible design options.

The key objective is to show that you can:

1. Generate multiple engineering design options
2. Evaluate them using engineering reasoning
3. Select a preferred solution using defensible trade-off analysis
4. Define a high-level architecture of the system
5. Identify subsystems and interfaces

D1 asks: *Why does this problem deserve an engineering solution?* D2 asks: *What engineering solution approaches could solve it, and why is one preferable?*

According to the module outcomes, Deliverable 2 must demonstrate that students can generate defensible engineering design options and evaluate them using engineering reasoning.

Core Components of Deliverable 2:

Deliverable 2 consists of five major elements:

1. Development of design options
2. Engineering trade-off analysis
3. High-level design of the selected option
4. Subsystem definition and interfaces
5. Subsystem integration at a high level

These components must form a logical engineering argument, not isolated sections.

1: Development of Design Options

What this means:

At this stage you must explore multiple possible engineering solutions to the problem identified in D1.

A critical principle of engineering design is: Engineers do not jump to a solution — they explore the solution space.

Therefore you must demonstrate that:

- Multiple plausible solution approaches exist – this is the first step.
- Each approach is technically feasible – you are capable of building all the solutions
- Each approach satisfies the user requirements from D1 – they can all satisfy the user requirements

Typical examples of design alternatives might include:

- Different sensing approaches (not 2 different specific sensors, like DHT 11 vs DHT 22). High level.
- Different control strategies
- Different system architectures
- Different communication technologies
- Different power systems

Important:

Each option must be technically justified, not simply listed. Show how all the possible approaches can be used. Discuss at least 3 solutions. Students must show creativity in exploring solution options and various engineering alternatives. You must show understanding of technology capabilities and limitations. Poor submissions typically present only one idea, present obvious solutions without analysis or jump directly to implementation details.

2. Engineering Trade-Off Analysis

What this means:

Engineering design is fundamentally about balancing trade-offs.

Once multiple design options are identified, you must compare them using engineering criteria. Typical criteria include:

- Technical performance
- Reliability
- Power consumption
- Cost
- Complexity
- Scalability
- Maintainability
- Environmental impact

The comparison must be structured and defensible. Typical methods include:

- Decision matrices
- Weighted scoring systems
- Engineering calculations
- Simulation comparisons

Example trade-off table:

Option	Accuracy	Power	Cost	Complexity	Score
GPS	High	High	Medium	Low	7
LoRa triangulation	Medium	Low	Low	Medium	8
Computer vision	High	High	High	High	5

Important:

This chapter needs evidence that you understand engineering constraints, can quantify trade-offs and that your decision making is transparent and justified.

The chosen solution must emerge from engineering reasoning, not preference.

3. High-Level Design of the Selected Option

What this means:

After evaluating the alternatives, the student must select one design approach and define its architecture.

This is where the design moves from conceptual options to system architecture. However, this is still high-level design, not detailed implementation and no device specifications or component numbers etc.

High-level design typically includes:

- System architecture diagram
- Functional block diagram
- Data flow diagrams

- User interfaces

This stage answers:

- What major components exist?
- What does each component do?
- How do they interact?

Important:

Do not get confused:

High-level design answers *What subsystems exist and what must they do?*

Detailed design (D3) answers *How exactly do those subsystems work?*

4. Subsystem Definitions and Interfaces

What this means:

A system is made of subsystems. Deliverable 2 requires identifying these subsystems and explaining their roles. Typical subsystem examples:

- Power subsystem
- Communication subsystem
- Sensing subsystem
- Processing subsystem
- User interface subsystem

For each subsystem the student must define:

- Purpose
- Inputs
- Outputs
- Dependencies

Example:

Subsystem	Function
Power system	Provides regulated power
Sensor module	Measures environmental data
Controller	Processes sensor information
Communication	Sends data to cloud

Interfaces define how subsystems interact. Clear subsystem interfaces are essential for systems engineering. Examples include:

- Electrical interfaces
- Communication protocols
- Data interfaces
- Mechanical interfaces
- User interfaces

5. Subsystem Integration (High Level)

What this means:

Finally, Deliverable 2 must demonstrate that the subsystems form a coherent system. This means explaining:



1. how subsystems interact
2. how data flows through the system
3. how the system achieves the user requirements

Integration must be described conceptually. Example integration explanation:

1. Sensors collect environmental data.
2. Controller processes measurements.
3. Communication subsystem transmits data to server.
4. User interface displays information.

Important:

This deliverable do not include PCB layouts, algorithms, code, detailed component choices. Those appear in Deliverable 3.

What Deliverable 2 must NOT contain

Students often make the mistake of moving too far ahead. Deliverable 2 should **not include**:

- Detailed circuit designs
- Component selection
- Software algorithms
- PCB layouts
- Implementation details
- Prototype results

Those belong to **D3 and D4**.

How Deliverable 1 will be assessed

According to the module outcomes, D2 is evaluated based on the following criteria:

- Development and justification of design options
- Engineering trade-off analysis
- High-level design of selected solution
- Subsystem definitions and interfaces
- High-level subsystem integration

The key focus of assessment is:

- engineering reasoning
- system thinking
- defensible decision-making

Marks are not awarded for:

- pretty diagrams
- large amounts of text
- superficial design descriptions

Marks are not awarded for:

- Writing style or formatting
- Document length
- Visual polish
- AI-generated text you cannot explain

If you cannot explain your design verbally, it will not be accepted as your own work.

